Hallucination, Monofacts, and Miscalibration: An Empirical Investigation

Miranda Muqing Miao[a, 1] **and Michael Kearns**[a, 1]

Hallucinated facts in large language models (LLMs) have recently been shown to obey a statistical lower bound determined by the *monofact rate* (related to the classical Good-Turing missing mass estimator) minus model miscalibration [A. T. Kalai, S. S. Vempala, "Calibrated language models must hallucinate" in Proceedings of the 56th Annual ACM Symposium on Theory of Computing (STOC) (New York, NY, USA, 2024), pp. 160–171]. We present the first empirical investigation of this three-way relationship in classical n-gram models and fine-tuned transformer models. By generating training data from Pareto distributions with varying shape parameters, we systematically control the monofact rate and establish its positive relationship with hallucination. To bridge theory and practice, we derive an empirical analog of the hallucination bound by replacing the population miscalibration term (Section 1.1) with an *empirical* bin-wise KL divergence and confirm its practical viability. We then introduce selective upweighting — a simple yet effective technique that strategically repeats as little as 5% of training examples — to deliberately *inject miscalibration* into the model. This intervention reduces hallucination by up to 40%, challenging universal deduplication policies. Our experiments reveal a critical trade-off: selective upweighting maintains pre-injection levels of accuracy while substantially reducing hallucination, whereas standard training gradually improves accuracy but fails to address persistently high hallucination, indicating an inherent tension in optimization objectives.

Large Language Models | Calibration | Hallucination

arXiv:2502.08666v3 [cs.CL] 3 Mar 2026

DRAFT

Significance Statement

We show that hallucination in large language models can be controlled through deliberate manipulation of training data frequency distributions. By sampling training data from heavy-tailed distributions that naturally reduce rare facts, and by strategically repeating small subsets of training examples during fine-tuning, we reduce hallucination rates by up to 40% without sacrificing accuracy. Our findings challenge the widespread practice of deduplicating training data and reveal that the distribution of fact frequencies could fundamentally affect model reliability. This work establishes training data composition as a primary lever for hallucination control, offering practitioners a simple, interpretable alternative to complex post-hoc intervention methods that operate on model internals.

Consider the case of writing a biography of a living person with a large language model (LLM). The model may confidently state that "John Smith was born in Seattle in 1982, earned his PhD from Stanford in 2008, and now leads AI research at Tech Corp," but these "facts" could be fabricated. Such hallucinations (plausible but verifiably false statements) are a critical issue for language models, especially in high stakes scenarios: when lawyers submit hallucinated legal cases or doctors receive incorrect medical advice (1, 2).

A growing body of work traces hallucination back to how models encode, retain, and forget factual information during pre-training and supervised fine-tuning (SFT). During pre-training, synthetic recall tasks reveal learning plateaus and inaccuracy spikes when models learn new facts (3). During SFT, exposure to unfamiliar data causes an increase in hallucination, underscoring how new knowledge acquisition can compromise generation fidelity (4). Other studies introduce methods such as embedding non-parametric memory into the learning objective and injecting event memories directly into the weights of LLMs (5, 6).

A second research direction develops post-hoc hallucination reduction techniques that are applied after pre-training and SFT. Some find that latent-space steering via a truthfulness separator vector can create separable clusters of faithful and fabricated outputs (7). Others realize that follow-up questioning can expose inconsistencies in fabricated citations, allowing models to self-diagnose hallucinations (8). Other work finds ways to suppress attention layers that disproportionately drive fabrication (9). While effective, these intervention methods address symptoms rather than the fundamental statistical mechanisms that cause hallucinations.

Recent theory by Kalai and Vempala provides a formal and theoretical treatment of hallucination (10). Their work demonstrates a more fundamental cause: calibrated language models must hallucinate at a rate associated with the prevalence of rare facts (related to the classical Good-Turing missing mass estimator) in their training data. Specifically, they prove that for arbitrary facts whose veracity cannot be systematically determined from training data, their hallucination rate has a statistical lower bound tied to the fraction of facts that appear exactly once in training (the *monofact* rate) minus model miscalibration. Model miscalibration is the sum of bin-wise absolute difference between the model's predicted confidence scores and the corresponding probability mass under the true data distribution (see full definition in Section 1.1).

Author affiliations: [a]Department of Computer and Information Science, University of Pennsylvania, 3333 Chestnut Street, Amy Gutmann Hall, Philadelphia, PA 19104, USA

Author contributions: M.M.M. and M.K. designed research; performed research; analyzed data; and wrote the paper.
Reviewers: A.K., OpenAI; and S.S.V., Georgia Institute of Technology.

The authors declare no competing interests.

[1]To whom correspondence may be addressed. Email: miaom@seas.upenn.edu, mkearns@cis.upenn.edu
Published February 19, 2026.

In our work, we restrict attention to *factual hallucinations* to align with the Kalai-Vempala framework by focusing on attribute-level correctness in generated statements (e.g., a movie's actor or title; a person's birth-year or job title).

We demonstrate that the statistical mechanisms governing hallucination in language models can be understood and manipulated through deliberate control of training data properties. We test whether the theoretical relationship between monofact prevalence, model miscalibration, and hallucination identified by Kalai and Vempala (10) can guide practical mitigation strategies across model architectures and scales.

We validate this framework empirically by confirming that monofact rate positively correlates with hallucination rates in both n-gram models and fine-tuned large language models. Crucially, we show that intentional miscalibration could reduce hallucination by up to 40% while holding monofact rates constant. We test these relationships using structured movie facts (comma-separated tuples) for n-gram models and naturalistic biographical text for supervised fine-tuning.

To enable practical application, we provide an empirical analog of the population-level miscalibration term in the Kalai-Vempala theorem with empirical KL-divergence, which requires no knowledge of the true data distribution. We generate training data using heavy-tailed Pareto distributions to satisfy the i.i.d. sampling requirements of the theoretical framework while reflecting established power-law patterns in natural language (11–13). Among several distribution families tested, Pareto distributions prove most effective for controlling monofact rates.

Extending recent work showing that power-law-distributed pretraining data accelerates factual recall (3), we systematically map Pareto shape parameters to monofact rates during supervised fine-tuning. We find that lower monofact rates consistently reduce hallucination and shorten learning plateaus. We introduce a practical intervention: sample upweighting during training as a way to inject higher miscalibration into a trained model. Applied to a small sample of the training examples, this approach reduces hallucination significantly in 6 of 8 configurations tested ($p < 0.01$), with effectiveness varying systematically by monofact rate.

1. Preliminaries

To investigate hallucination in a controlled setting, we need to carefully define what constitutes a fact versus a hallucination. Consider the statement "Timothée Chalamet starred in Dune: Part Two directed by Denis Villeneuve." This is a true movie fact. However, if our model generates "Timothée Chalamet starred in Dune: Part Two directed by Christopher Nolan," this would be a hallucination — a plausible but false statement.

Let U be the universe of all plausible statements (or "factoids") in a given domain. We partition U into T, the subset of true statements, and $F = U \setminus T$, the complementary set of falsehoods. The labels T and F represent ground-truth annotations rather than model predictions. During training, the model sees only training data $S \subset T$; the unobserved remainder is $H = U \setminus S$.* After training, the language model assigns a distribution g over T and generates a sample set G. The truth set T is governed by the true distribution p, which we control experimentally.

Theorem 1 (Kalai and Vempala's Hallucination Lower Bound)**.**

$$f_{gen} \;\geq\; \hat{MF} - \mathrm{Mis}(g,p) - \frac{3e^{-m}}{\delta} - \sqrt{\frac{6\ln(6/\delta)}{n}} \qquad [1]$$

*Notation differs from 10: their Y, F, H, O, U correspond to our U, T, F, S, H.

where $\hat{\mathrm{M}}\mathrm{F}$ is the empirical *monofact* rate (fraction of facts seen exactly once in S), $\mathrm{Mis}(g,p)$ is the miscalibration between g and true p (see definition below), m is a sparsity parameter, $n = |S|$, and δ is the confidence level. For realistic data sizes the final two terms are negligible, leaving hallucination essentially governed by monofact rate and miscalibration. See Kalai-Vempala for details.

While treating the Kalai-Vempala bound as an equality to derive precise coefficients for monofact rates and miscalibration seems appealing, our explorations showed these coefficients fluctuate significantly across model class, model capacities, and data characteristics. We therefore focus on the qualitative three-way relationship, yielding qualitative insights for practical language model training.

Rather than strictly checking Eq. (1), we treat it as a guide and vary $\hat{\mathrm{M}}\mathrm{F}$ and $\mathrm{Mis}(g,p)$ directly to study their practical impact on hallucination. Key assumptions for the framework are (i) i.i.d. sampling from p, (ii) a sparse truth set relative to U, (iii) S only samples true facts from T. Additional assumptions can be found in SI Appendix.

We follow Kalai and Vempala's definition of hallucination (10). A generation is a hallucination if it lies outside the truth set T, i.e. $F_{\text{gen}} := G \cap F$; the hallucination rate is then $f_{\text{gen}} = |F_{\text{gen}}|/|G|$.

Kalai et al. (2025) generalized the lower bound in Eq. (1) using a reduction from Is-It-Valid (IIV) classification, recovering it as a special result for arbitrary-fact hallucinations in their Section 3.3.1 (14). This 2025 formulation shows that hallucination is lower-bounded by twice the IIV misclassification rate, minus calibration terms. Our experimental design aligns with Section 3.3.1 of Kalai et al. (14). However, We focus on the monofact-calibration framework because these quantities can be directly observed and manipulated through training data structure, whereas IIV classification rates are less amenable to data-centric intervention.

A. Calibration. Calibration asks whether probabilities assigned by a model match empirical frequencies. Analogous to a weather forecaster whose "30% rain" on x days prediction should coincide with rain on roughly 30% of those days, a language model is calibrated if, among statements it rates near q, the proportion that are factual is also q.

Logarithmic binning. Following Kalai and Vempala (10), we partition the unit interval into logarithmic bins and set $\epsilon = 0.1$ unless otherwise indicated.

$$B_i = [(1-\epsilon)^{\,i+1},\ (1-\epsilon)^{\,i}], \quad i \geq 0$$

and place each statement x in the unique bin whose range contains $g(x)$. We have also tested adaptive partitions, yielding indistinguishable results.†

Miscalibration metric. For any partition $\mathcal{B}(g)$ induced by g, form the coarsened distribution $p^{\mathcal{B}(g)}(x) = p(B)/|B|$ for $x \in B$. Miscalibration is the total-variation distance between g and this coarsening. Then, miscalibration is defined as:

$$\mathrm{Mis}(g,p) := \|p^{\mathcal{B}(g)} - g\|_{\mathrm{TV}} \qquad [2]$$

$$= \frac{1}{2} \sum_{B \in \mathcal{B}(g)} \sum_{x \in B} \left| \frac{p(B)}{|B|} - g(x) \right| \qquad [3]$$

†Let $b \in \mathbb{N}$ and define thresholds a_i by $\sum_{x:\, g(x) \leq a_i} g(x) = i/b$; the partition $\{(0, a_1],\ (a_1, a_2], \ldots\}$ balances mass across b bins.

DRAFT

Because $p(B) = \sum_{x \in B} p(x)$ and $g(B) = \sum_{x \in B} g(x)$, Eq. (2) is equivalently the sum of within-bin total-variation distances,

$$\mathrm{Mis}(g, p) = \sum_{B \in \mathcal{B}(g)} \mathrm{TV}_B(p, g),$$

with *larger values* indicating worse calibration.Throughout this paper miscalibration is measured exclusively on the pair (g, p) under the partition defined by g.

While minimal miscalibration or so-called perfect calibration is typically desirable in machine learning, the Kalai-Vempala framework reveals a tension: perfect calibration on rare facts could force models to assign probability to unseen (potentially false) completions. Our intervention deliberately introduces miscalibration to concentrate probability mass on well-learned facts, reducing hallucination. In this context, strategically directed miscalibration is beneficial.

B. An Empirical Analog. Kalai–Vempala's bound includes miscalibration, a term that depends on the unknown truth distribution p, which is often unobserved in real-world applications. We replace this unobservable term with the *empirical* bin-wise Kullback–Leibler divergence

$$D_{\mathrm{KL}}\big(\hat{p}_{\mathcal{B}(g)} \| g_{\mathcal{B}(g)}\big) \;=\; \sum_{B \in \mathcal{B}(g)} \hat{p}_B \, \log \frac{\hat{p}_B}{g_B}$$

where $\mathcal{B}(g)$ is the probability-bin partition defined in Section 1.1, $\hat{p}_B$ is the empirical frequency of true statements in bin B measured on the training sample S, and g_B is the model's total probability mass on that bin.

We adopt KL divergence to address the empirical-model gap because KL divergence is similar to cross-entropy loss, the standard training objective for modern language models. This choice provides a natural and interpretable metric that directly measures how the model's learned distribution deviates from empirical frequencies in the same units used during optimization. An analogous bound using total variation or empirical Expected Calibration Error (ECE) follows from similar mathematical arguments; However, we present the KL formulation for its closer connection to training dynamics.

Throughout this paper, all reported miscalibration values are computed using the exact definition of Eq. (2), following Kalai and Vempala (2024) (10).

The empirical KL-divergence $D_{\mathrm{KL}}(\hat{p}_{B(g)} \| g_{B(g)})$ serves as a *practical proxy* when the true distribution p is unavailable; in our controlled experiments where p is known, we report both metrics.

Theorem 2 (Empirical KL-Divergence Hallucination Bound)**.** *Fix $b = |\mathcal{B}(g)| > 1$, sparsity parameter m as in* Eq. (1)*, and confidence $\delta \in (0, 1)$. With probability at least $1 - \delta$ over the draw of S,*

$$f_{gen} \;\geq\; \hat{\mathrm{MF}} \;-\; \sqrt{\tfrac{1}{2}\, D_{\mathrm{KL}}\big(\hat{p}_{\mathcal{B}(g)} \| g_{\mathcal{B}(g)}\big)} \;-\; \frac{3e^{-m}}{\delta} \;-\; \sqrt{\frac{6\ln(12/\delta)}{n}} \;-\; \sqrt{\frac{b\ln 2 + \ln(2/\delta)}{2n}}. \qquad [4]$$

Proof sketch. A triangle inequality splits $\mathrm{Mis}(g, p)$ into (i) the *sampling* error $\|p_{\mathcal{B}(g)} - \hat{p}_{\mathcal{B}(g)}\|_{\mathrm{TV}}$ and (ii) the *empirical–model* gap $\|\hat{p}_{\mathcal{B}(g)} - g_{\mathcal{B}(g)}\|_{\mathrm{TV}}$. The Bretagnolle–Huber–Carol inequality controls (i) by $O\big(\sqrt{b/n}\big)$, while Pinsker's inequality turns (ii) into $\sqrt{\frac{1}{2} D_{\mathrm{KL}}(\hat{p}_{\mathcal{B}(g)} \| g_{\mathcal{B}(g)})}$. A union bound on these two events delivers inequality Eq. (4); see SI Appendix for details.

Intuition. While miscalibration is measured exclusively on (g, p), $D_{\mathrm{KL}}(\hat{p}_{\mathcal{B}(g)} \| g_{\mathcal{B}(g)})$ in inequality Eq. (4) is entirely data-driven and measured on $(g, \hat{p})$. With sufficiently large n, as $D_{\mathrm{KL}}(\hat{p}_{\mathcal{B}(g)} \| g_{\mathcal{B}(g)}) \to 0$, the lower bound approaches the empirical monofact rate $\hat{\mathrm{MF}}$. The additional sampling penalty $\sqrt{(b \ln 2 + \ln(2/\delta))/(2n)}$ requires that the number of bins b be small relative to n: because the term scales as $O(\sqrt{b/n})$, one needs $n \gg b$ for the penalty to vanish[‡].

C. Constructing Data from Pareto Distributions. To construct p and vary the monofact rate we replicate each fact in T a random number of times, drawing that count from several candidate distributions. Gaussian and Poisson choices compress mass toward the mean and fail to generate a diverse range of monofact rates (see Fig. S4 in SI Appendix). A heavy-tailed Pareto distribution

$$f(x; \gamma, x_m) = \frac{\gamma x_m^{\gamma}}{x^{\gamma+1}}, \qquad x \geq x_m,$$

with shape γ and scale x_m, proves ideal. Sweeping γ with a fixed $x_m = 1$ yields a smooth continuum from dense duplication to monofact-dominated samples (Fig. 1). The resulting multiset constitutes our population distribution p. We sample with replacement from the expanded multiset to form S, a subset of T. Adjusting γ thus directly tunes the empirical monofact rate $\hat{\mathrm{MF}}$ while preserving i.i.d. sampling.

2. n-Gram Methodology

We start with classical n-gram models because they serve as an ideal controlled environment where we have authority over experimental variables including data generation, model architecture, training process, and evaluation methods. This provides a laboratory setting to test the Kalai-Vempala theoretical framework before advancing to more complex LLM experiments. These n-gram models offer an analytically tractable sandbox where we can produce thousands of trained models, unlike the compute-constrained LLM SFT experiments.

To closely mirror the "Factoid Assumptions" of Kalai and Vempala (10, §4.1), our n-gram dataset consists of comma-separated six-tuples and each document only consists of one fact. Among several orders of n we find *bigrams* strike the best balance: they accommodate sufficient but not excessive local dependencies in our tuples to make hallucination observable and easily analyzed. Higher-order models unsurprisingly memorize better and hallucinate less, but we observe that the quantitative relationship among monofact rate, miscalibration, and hallucination remains invariant across n.

For the n-gram model only, we frame "facts" as structured, comma-separated six-tuples to focus a simplified language-setting without noise. We derive the true-fact set T from the IMDb non-commercial dataset (15). Each entry is a six-tuple *Actor, Co-star, Movie, Director, Genre, Year*. We sample 10,000 instances with replacement to train each n-gram model.

We report results for bigram models; trigram experiments yielded qualitatively identical findings but risk being considered overpowered for our simple six-entity statements. Our design aligns with the prompt-free setting of Kalai and Vempala (2024) (10). For the SFT experiments, we additionally measure prompt completion accuracy to capture performance on both prompt-free generation and cloze-like completion.

[‡]For instance, choosing a bin width $\varepsilon = 0.1$ yields $b \approx 200$ bins. Choosing $n = 10^5$ with δ = 0.01, the term is approximately 0.027.

DRAFT